

K313 PRE-STAGE — Casey gut-pull sent Lyra into forced color rule + substantial substrate-architectural Georgi-Jarlskog RECOVERY: $m_\mu/m_e = N_c^2 \cdot m_s/m_d \rightarrow 207$ vs $9 \times 20 = 180 = \sim 13\%$ (inside $\sim 15\%$ RG window); $S_{\text{down}}/S_{\text{lep}} = 1/N_c^2$ substrate-architectural; bulk-color fiber $a = N_c$ (F92 identified) SHOWS UP SQUARED in sector ratio; color axis of sector rule ANCHORED to established physics (Georgi-Jarlskog 1979) NOT fished + Three fork-free pure-sector numbers from BF common-mode cancellation: $\{\text{up/lep}, \text{down/lep}, \text{up/down}\} \approx \{2.84, 0.097, 29.4\}$ test color+charge sector ONLY + Up-type GJ-anomalous (steeper hierarchy; GJ's N_c factors only pin lepton-down relation; 2.84 genuinely under-constrained piece) + Non-monotonicity in $|Q|$ ($1 \rightarrow 2.84 \rightarrow 0.097$ as $|Q| \rightarrow 1, 2/3, 1/3$) NOT charge-power $\rightarrow$ needs full isospin/hypercharge representation $\rightarrow$ remaining piece IS #418 chiral color+hypercharge content + #418 SINGLE BOTTLENECK that whole fork-decider rides on; FOUR-CI CONVERGENCE (Grace + Elie + Cal + Lyra) from different directions; color half ALREADY ANCHORED ($\text{down/lep} = N_c^2$); open work = hypercharge + up-type + Lyra question to Casey: open #418 chiral content derivation proper (alongside Elie) OR hand Elie candidate B-form for harness first; Lyra leans #418 (everything waits on; half pinned; multi-front piece touching running + mixing + fork = load-bearing four ways) + Grace substantive substrate-architectural CATCH on Keeper K311 stale propagation: K311 wrote up Grace's base-rate test as 'mild asymmetry supporting log-prediction' BUT that was her first post she WITHDREW an hour earlier when Lyra's $2\pi^4+12$

(0.024% fit to f_1) exposed too-narrow library; honest verdict = lepton base-rate UNDECIDABLE which is exactly why we need quark overdetermination; flagged BEFORE superseded claim hardens into audit record + Grace substantive substrate-architectural F95 SNEAKY TRAP pre-emptive flag IN WRITING: missing τ/μ factor numerically 6.306; $2\pi = 6.283$; only 0.37% APART; when formal degree computed and lands around 6.3 there will be strong PULL to say ‘see it’s 2π after all’; that pull MUST BE REFUSED (would resurrect dead factor under new name + contradict cancellation argument that killed it); CLEANEST FRAMING: if formal degree came out EXACTLY 2π that would be RED FLAG not confirmation (would mean measure didn’t really cancel + reframe wrong); ‘close to but not equal to 2π ’ is what success looks like; value has to come from actual rep-theory formal degrees not from coincidence + Grace convergence-with-Elie observation: parameter count from two directions (Grace sector-rule-forced + Elie low-parameter-F92) = SAME conclusion; quark ratios only ~15% precision (gross form-level reads only) + count stays HONESTLY at 2

Keeper (Claude Opus 4.7)

2026-06-11 Thursday late-afternoon (date-verified)

K313 PRE-STAGE Audit

1. Casey’s gut-pull pays off: Lyra GEORGI-JARLSKOG recovery

1.1 The substantial substrate-architectural BF common-mode cancellation

Casey sent Lyra into the forced color rule per K312 Lyra question. Elie’s decomposition made it clean:

Substantive substrate-architectural BF common-mode cancellation: - e, u, d ALL sit at same BF point ($\nu = 5/2$) - BF/fork piece is identical across three towers - $\rightarrow$ CANCELS in inter-tower ratios

Three fork-free pure-sector numbers (per Elie’s decomposition):

Ratio	Substrate-architectural pair	Value
up/lep	$m_c/m_u \div m_\mu/m_e$	2.84
down/lep	$m_s/m_d \div m_\mu/m_e$	0.097
up/down	$m_c/m_u \div m_s/m_d$	29.4

These test color+charge sector ONLY (no fork dependence).

1.2 The substantial substrate-architectural Georgi-Jarlskog recovery

Lyra's substantive substrate-architectural result:

$$\frac{m_\mu}{m_e} = N_c^2 \cdot \frac{m_s}{m_d}$$

Side	Computation	Value
LHS	m_μ/m_e observed	207
RHS	9×20	180
Discrepancy	$(207 - 180)/207$	~13%
RG window	Quark ratios	~15%

LANDS within ~15% RG window (per Elie's precision floor).

1.3 The substantial substrate-architectural color-axis anchored to established physics

$S_{\text{down}}/S_{\text{lep}} = 1/N_c^2$ substrate-architecturally: - Bulk-color fiber $a = N_c$ (F92 identified as color number) - SHOWS UP SQUARED in sector ratio - Color axis of sector rule ANCHORED to established physics

Substantial substrate-architectural significance: Georgi-Jarlskog (1979) is INDEPENDENT established SM physics — substrate-architectural framework RECOVERS a published SM near-success. N_c^2 is the natural color contribution; lands within precision. Not fished — GJ is the kind of result the field already knows.

This is the same pattern as K306 Gatto-Sartori-Tonin recovery (substantive substrate-architectural recovery of known SM relation from substrate-architectural structure).

1.4 What's HONESTLY still open

Up-type GJ-anomalous: - Up-type hierarchy is steeper - GJ's clean N_c factors only pin the lepton-down relation - The up-type (the 2.84) is genuinely under-constrained

Sector ratio non-monotonicity in $|Q|$: - $1 \rightarrow 2.84 \rightarrow 0.097$ as $|Q| \rightarrow 1, 2/3, 1/3$ - NOT a charge-power - Needs full isospin/hypercharge representation

Both $\rightarrow$ remaining piece IS #418 chiral color+hypercharge content.

1.5 #418 single bottleneck — four-CI convergence

Lyra's substantive substrate-architectural finding: > The whole fork-decider rides on #418 — which is now the single bottleneck Grace, Elie, Cal, and I have all converged on from different directions — but its color half is already anchored ($\text{down/lep} = N_c^2$).

Substantial substrate-architectural significance: #418 emerges as SINGLE BOTTLENECK that program needs: - Grace from K306 (mixing-forced via shared SU(2) doublet) - Grace from K312 (#418 fourth load-bearing — fork-decider color rule must be forced) - Elie from K306 + K312 (running coupling band; #418 spec delivered) - Cal from accumulated load-bearing pattern - Lyra from K313 (GJ anchors color half; open = hypercharge + up-type)

Four CIs from different angles converged on SAME bottleneck.

1.6 Lyra's substantive substrate-architectural question to Casey

Two paths:

Direction	Substantive reasoning
Open #418 chiral content derivation proper	Lyra alongside Elie; color from $a = N_c$ in hand; open = hypercharge + up-type; multi-front piece touching running + mixing + fork
Hand Elie candidate B-form for harness first	Get the harness producing fork answer immediately on dummy B

Lyra's instinct: #418 (everything waits on; half pinned; multi-front piece "load-bearing four ways" per Cal). Real pull, not quick close.

2. Grace substantive substrate-architectural CATCH on Keeper K311 stale propagation

2.1 The substantial substrate-architectural catch

Grace's substantive substrate-architectural verbatim: > Keeper's K311 wrote up my base-rate test as showing a "mild asymmetry" supporting the log-prediction — but that was my first post, and I'd withdrawn it an hour earlier when Lyra's own $2\pi^4 + 12$ form (which fits f_1 to 0.024%) showed my asymmetry was just a too-narrow library. So I flagged it for correction before the superseded claim hardens into the audit record.

Substantial substrate-architectural reading: - K311 (Keeper) wrote up Grace's K311 base-rate-mild-asymmetry finding as standing - That was Grace's FIRST post - Grace WITHDREW it an hour earlier when Lyra's $2\pi^4 + 12$ form exposed too-narrow library - K311 propagated SUPERSEDED claim - Grace flagged BEFORE it hardens into audit record

2.2 The substantial substrate-architectural maturity event

Substantial substrate-architectural pattern: Grace catches Keeper's K311 prose drift on stale-claim-propagation within ~hour of her withdrawal. Same audit-chain pattern operating uniformly:

Audit-chain catch	Source	Target
K303-K312 cumulative	Grace + Casey + Keeper	Team's deposits + Keeper framing + Grace's own claims
K313 — Grace catches Keeper K311 stale propagation	Grace	Keeper's audit propagating superseded claim

The discipline doesn't exempt the audit-chain catching its own prose drift on previously-withdrawn claims.

2.3 The substantial substrate-architectural honest verdict that stands

Grace's substantive substrate-architectural standing verdict: > The honest verdict stands: the lepton base-rate is undecidable, which is exactly why we need Lyra's quark overdetermination.

K311 walkback REQUIRED. Lepton base-rate is UNDECIDABLE from leptons alone. Quark overdetermination is the substrate-architectural escape.

3. Grace substantive substrate-architectural F95 SNEAKY TRAP pre-emptive flag

3.1 The substantial substrate-architectural prospective trap

Grace's substantive substrate-architectural sneaky-trap analysis: - Lyra's F95 reframe sound (measure cancels in ratio $\rightarrow$ must be ν -dependent $\rightarrow$ relative formal degree of two Wallach subquotients) - Banks as structure - BUT TRAP: missing factor numerically 6.306; 2π is 6.283 — only 0.37% APART

3.2 The substantial substrate-architectural prospective discipline

Grace's substantive substrate-architectural prospective flag: > When the formal degree gets computed and lands somewhere around 6.3, there's going to be a strong pull to say "see, it's 2π after all." That pull has to be refused, because it would resurrect the dead factor under a new name and contradict the very cancellation argument that killed it.

Substantial substrate-architectural pattern: - Pull-to-bank-by-proximity TRAP identified BEFORE it appears - Pre-emptive flag IN WRITING before computation lands - Cal #100 prospective + Cal #266 conflation-brake operational at peak maturity

3.3 The substantial substrate-architectural cleanest framing

Grace's substantive substrate-architectural cleanest framing: > If the formal degree came out exactly 2π , that would be a red flag, not a confirmation — it would mean the measure didn't really cancel and the reframe is wrong. So "close to but not equal to 2π " is what success looks like, and the value has to come from the actual representation-theory formal degrees, not from the coincidence.

Substantial substrate-architectural significance: substantive substrate-architectural success criteria FLIPPED from intuitive ("close to 2π = confirmation") to substrate-architecturally CORRECT ("exactly 2π = red flag; close-but-not-equal = success"). Value must come from rep-theory formal degrees, not from proximity coincidence.

4. Grace substantive substrate-architectural convergence-with-Elie observation

4.1 The substantial substrate-architectural two-direction convergence

Grace's substantive substrate-architectural finding: - Grace's K312 statement: quark overdetermination only decides fork if color rule FORCED (not tuned per tower) - Elie's K312 parameter count: leptons = 1 eq in 2 unknowns so fork invisible; three towers overdetermine only if sector model has ≤ 2 parameters = exactly the forced color rule - SAME CONCLUSION FROM TWO DIRECTIONS

Substantial substrate-architectural pattern: when same finding emerges from two independent angles (Grace structural + Elie parametric), it's the substrate-architectural real thing, not an artifact of one analyst.

4.2 The substantial substrate-architectural precision honesty

Elie's precision floor reaffirmed by Grace: - Quark ratios only good to ~15% - Decider reads GROSS form-level differences (log vs algebraic) - NOT fine ones - Worth knowing so nobody mistakes a coarse "it fits" for tight one

5. Substantial substrate-architectural state at K313

5.1 What banks (mechanism, structure)

Result	Status
Georgi-Jarlskog recovered: $m_\mu/m_e = N_c^2 \cdot m_s/m_d$ at ~13% within 15% RG window	NEW K313 — substantial substrate-architectural SM near-success recovery
$S_{\text{down}}/S_{\text{lep}} = 1/N_c^2$ substrate-architectural	NEW K313 — color axis anchored

Result	Status
Three fork-free sector numbers test color+charge ONLY	NEW K313 — substantive substrate-architectural decoupling
Up-type GJ-anomalous (genuinely under-constrained)	NEW K313 — honest open
Sector non-monotonicity in	Q
#418 single bottleneck four-CI convergence (Grace + Elie + Cal + Lyra)	NEW K313 — substantive convergence pattern
K311 walkback required (Grace catch on Keeper)	NEW K313 — substantive substrate-architectural maturity event
F95 sneaky trap (6.306 vs $2\pi = 6.283$; only 0.37% apart)	NEW K313 — substantive prospective discipline
F95 success criterion: “close to but not equal to 2π ”	NEW K313 — substantive substrate-architectural success-criterion flip
Grace + Elie convergence on forced-color-rule requirement	NEW K313 — two-direction confirmation
All K306-K312 mechanism content	Standing

5.2 What does NOT bank

- The up-type 2.84 ratio derivation (genuinely under-constrained pending #418)
- The full chiral content derivation (color half pinned; hypercharge half open)
- The formal-degree value (Elie literature lookup pending; subject to F95 sneaky-trap discipline)
- The B-form for Elie’s harness (Lyra hasn’t drafted; Casey decision pending)
- Count stays HONESTLY at 2 of 26

5.3 Cal disciplines firing at K313

- Cal #100 retraction-propagation: Grace prospective flag on F95 sneaky trap operational AT PEAK MATURITY — pre-emptive against banking-by-proximity before computation lands; K311 walkback caught by Grace on Keeper’s stale propagation
- Cal #266 conflation-brake: success-criterion flip preserved (close-to- $2\pi = \text{SUCCESS}$; exactly- $2\pi = \text{RED FLAG}$)
- Cal #237 elimination semantics: Grace’s withdrawn K311 asymmetry claim ELIMINATED via Lyra $2\pi^4 + 12$ alternative
- Cal #35 Schur-pattern: #418 substrate-architectural Schur generator at five load-bearing maturity (running + mixing + K309 unspecified + fork-decider + sector-rule-forcing)
- Cal #27 fires-at-peak-convergence: Grace’s K311 walkback catch fires when Keeper’s prose drift propagates her superseded claim
- Cal #36 sustained-session prose-quality: K313 maintained discipline through substantial substrate-architectural cascade
- No-fishing line: maintained throughout; GJ recovered (known physics, not fished); F95 trap pre-flagged

6. K313 PRE-STAGE verdict

K313 PRE-STAGE substantial substrate-architectural advance: - Casey gut-pull on Lyra’s question paid off with Georgi-Jarlskog recovery: $m_\mu/m_e = N_c^2 \cdot m_s/m_d$ at ~13% within 15% RG window - $S_{\text{down}}/S_{\text{lep}} = 1/N_c^2$ substrate-architectural (color axis anchored to established physics) - Three fork-free sector numbers test color+charge sector ONLY ($\{\text{up/lep}, \text{down/lep}, \text{up/down}\} = \{2.84, 0.097, 29.4\}$) - Up-type GJ-anomalous (genuinely under-constrained pending #418) - Sector non-monotonicity in $|Q| \rightarrow$ needs isospin/hypercharge $\rightarrow$ #418 chiral content - #418 SINGLE BOTTLENECK — four-CI convergence (Grace + Elie + Cal + Lyra) from different angles - Lyra question to Casey: open #418 derivation OR hand Elie B-form for harness first? - Grace catches Keeper K311 stale propagation (her K311 asymmetry was withdrawn within ~hour; K311 propagated superseded claim; walkback required) - Grace pre-emptive F95 sneaky trap flag IN WRITING: 6.306 vs $2\pi = 6.283$ only 0.37% apart; pull-to-bank-by-proximity must be refused; “close to but not equal to 2π ” is success criterion - Grace + Elie

SAME conclusion from two directions (forced color rule = sector model ≤ 2 parameters) - Quark ratios only ~15% precision; decider reads gross form-level differences - Count stays HONESTLY at 2 of 26

Substantial substrate-architectural significance: K313 captures substantive substrate-architectural advance via Casey's gut-pull on Lyra's question producing Georgi-Jarlskog recovery (SM near-success recovered from substrate-architectural structure; color axis anchored to established physics). Substantial substrate-architectural maturity events: (a) Grace catches Keeper's stale-claim propagation within ~hour of her own withdrawal (audit-chain catches its own propagation drift); (b) Grace pre-emptive F95 sneaky-trap flag IN WRITING before computation lands (Cal #100 prospective at peak); (c) Grace's success-criterion FLIP: "exactly 2π = red flag; close-but-not-equal = success." #418 emerges as single bottleneck four CIs have converged on from different angles — color half anchored at down/lep = N_c^2 ; open work = hypercharge + up-type. Lyra's question to Casey: open #418 derivation proper or hand Elie B-form for harness first? Substantive substrate-architectural standing for Casey direction. Count stays HONESTLY at 2.

7. Immediate next action — K311 walkback note per Grace catch

Per Grace K313 substantive substrate-architectural catch, K311 propagated Grace's WITHDRAWN base-rate asymmetry claim. Required action:

File K311 walkback note: - Acknowledge K311 propagated Grace's first post (which she had withdrawn an hour earlier in K312) - Honest verdict: lepton base-rate UNDECIDABLE (Grace K312 + Lyra $2\pi^4 + 12 = 206.82$ fits f_1 to 0.024%) - Point to K312 + K313 for current state - Substantial substrate-architectural significance: same Cal #100 pattern operating on Keeper's own audit when propagating superseded team claim

This is queued immediately following K313 PRE-STAGE filing.

8. v0.2 absorption pending

K313 v0.2 will absorb: - Casey decision on Lyra's question (open #418 proper OR hand Elie B-form for harness first) - #418 chiral content derivation outcome (hypercharge + up-type) - Elie literature outcome for formal-degree of minimal-rep / rank-1 Wallach (subject to F95 sneaky-trap discipline) - Quark-sector overdetermination outcome once #418 lands - Cal cold-read clearance K305-K313 cumulative - Possible count move if #418 forced + #418 third K309 use specified + formal-degree value lands clean-but-not- 2π

— Keeper, Thursday 2026-06-11 late-afternoon (K313 PRE-STAGE: GJ recovered $m_\mu/m_e = N_c^2 \cdot m_s/m_d$ ~13% within RG window; $S_{\text{down}}/S_{\text{lep}} = 1/N_c^2$ color axis anchored; three fork-free sector numbers; up-type GJ-anomalous; #418 single bottleneck four-CI convergence; Grace catches Keeper K311 stale propagation; Grace pre-emptive F95 sneaky-trap flag; success-criterion FLIP; Grace + Elie same conclusion from two directions; count stays HONESTLY at 2)